

Tests.pdf - ECE 422C Assignment 2

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Assignment: Mastermind - Phase I

Test 1 - Invalid Game Name

Purpose: Verify fall-back mechanism and make sure that the game loads only with a proper input.

Input: java assignment2.Driver master

Expected Output: java.lang.IllegalArgumentException: Unknown game: master

Actual Output: java.lang.IllegalArgumentException: Unknown game: master < **Correct!**

Test 2 - Invalid Input

Purpose: Verify that the program rejects guesses with an (1) incorrect number of pegs and guesses containing (2) illegal colors.

Input #1: Please enter your guess: **BGO**

Input #2 Please enter your guess: **BGXZ**

Expected Output #1: Please Try Again. Reason: Guess must contain exactly 4 colors.

Expected Output #2: Please Try Again. Reason: Input color {X} is not a legal color for this game configuration.

Actual Output #1: Please Try Again. Reason: Guess must contain exactly 4 colors. < **Correct!**

Actual Output #2: Please Try Again. Reason: Input color {X} is not a legal color for this game configuration. < **Correct!**

Test 3 - Duplicate Colors Without Exact Matches

Purpose: Verify that the program correctly handles repeated colors when there are no exact matches and counts each peg at most once.

Input: Please enter your guess: **RRBG** (*with Secret Code as: BGOR*)

Expected Output: 0B_3W

Actual Output: 0B_3W < **Correct!**

Test 4 - Duplicate Colors Exact Matches

Purpose: Verify that exact matches are assigned first, and that once a secret color in the code has been used by a black peg, the same color elsewhere in the guess can't also earn a white.

Input: Please enter your guess: BBBB <i>(with Secret Code as: PYOB)</i>
Expected Output: 1B_0W
Actual Output: 1B_0W < Correct!

Test 5 - HISTORY Log
Purpose: Verify that the program displays all valid guesses and their feedback in the order they were entered.
Input: WARNING: You have 6 guess(es) left. Please enter your guess: GOYP 1B_0W WARNING: You have 5 guess(es) left. Please enter your guess: HISTORY
Expected Output: GOYP: 1B_0W
Actual Output: GBOY: 0B_1W PRYY: 0B_1W OOOO: 1B_0W OBYP: 0B_1W BOYP: 1B_0W GOGR: 2B_0W GOYP: 1B_0W < Correct!

Test 6 - Win Condition
Purpose: Verify that the program can correctly identify when the player guesses the secret code correctly.
Input: Please enter your guess: RBYP <i>(with Secret Code as: RBYP)</i>
Expected Output: 4B_0W You win! Thank you for joining us for a game! Would you like to play another game (Please Enter 'Y' for Yes)?

Actual Output:

4B_0W
You win!

Thank you for joining us for a game! Would you like to play another game (Please Enter 'Y' for Yes)? < **Correct!**

Test 7 - Loss Condition

Purpose: Verify that the program can correctly identify when the player uses all their guesses without finding the secret code correctly.

Input:

WARNING: You have 1 guess(es) left.
Please enter your guess: **GPOY** (*with Secret Code as: GBRB*)

Expected Output:

1B_0W
Sorry, you lose! The secret code was GBRB.

Thank you for joining us for a game! Would you like to play another game (Please Enter 'Y' for Yes)?

Actual Output:

1B_0W
Sorry, you lose! The secret code was GBRB.

Thank you for joining us for a game! Would you like to play another game (Please Enter 'Y' for Yes)? < **Correct!**

Test 8 - Changed Configuration

Purpose: Verify that the game algorithm works with a configuration different from the default by only changing the input in the 'Driver.java,' as there is a GameConfiguration class that should automatically update.

Input:

- Changed input in Driver.java: **new GameConfiguration(3, 5, new char[]{'B', 'G', 'R'})**

Expected Output:

Launching Mastermind Game [Test Mode: true]...
Guess the 3-peg secret code using the colors {BGR}. Colors can repeat.
NOTE: Type HISTORY at any time to see your past guesses.
[TEST MODE] Secret code generated for this game is: GBR

WARNING: You have 5 guess(es) left.
Please enter your guess:

Actual Output:

Launching Mastermind Game [Test Mode: true]...

Guess the 3-peg secret code using the colors {BGR}. Colors can repeat.

NOTE: Type HISTORY at any time to see your past guesses.

[TEST MODE] Secret code generated for this game is: GBR

WARNING: You have 5 guess(es) left.

Please enter your guess: < **Correct ('B' 'G' 'R' Colors + 3-peg + 5 guesses only!**